15/01/2021

CoE-XII

Aakash

Medical | IIT-JEE | Foundations

Regd. Office :Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

NCERT BOOSTER TEST SERIES

Test - 09

Topics covered in various subjects:

Physics :	Ray Optics, Wave Optics, Dual Nature of Radiation and Matter
Chemistry :	Haloalkanes and Haloarenes, Alcohols, Phenols and Ethers, Aldehydes and Ketones and Carboxylic Acids
Botany :	Complete syllabus of XI and XII
Zoology :	Complete syllabus of XI and XII

PHYSICS

1. Read following statements **A** and **B** carefully and choose the option which represents the correct answer

   Statement A: In order to explain the phenomena of refraction, Newton postulated that the speed of the corpuscles was greater in water or glass than in air.

   Statement B: In corpuscular picture of refraction particle of light incident from a rarer to denser medium experience a force of attraction normal to the surface. This results in the normal component of the velocity increases but the component along the surface is unchanged.

   (1) Only statement (A) is correct
   (2) Only statement (B) is correct
   (3) Both the statements (A) and (B) are correct
   (4) Neither statement (A) nor (B) is correct

2. Light can be considered to travel in a straight line because
   (1) It is reflected by obstacles
   (2) Its velocity is very large
   (3) It is refracted by obstacles
   (4) Its wavelength is very small compared to the size of obstacles

3. The earth takes 24 h to rotate once about its axis. How much time does the sun take to shift by 3° when viewed from the earth, is
   (1) 2 min (2) 12 min
   (3) 5 min (4) 4 min

4. Read following statements **A** and **B** carefully and choose the option which represents the correct answer

   Statement A: The real image of an object placed between f and 2f from convex lens can be seen on a screen placed at the image location. If the screen is removed, the image is still there.

   Statement B: Image formation needs regular reflection or refraction. In the principle, all rays from a given point should reach the same image point. This is the reason why we do not see our image by an irregular reflecting object.

   (1) Only statement (A) is correct
   (2) Only statement (B) is correct
   (3) Both the statements (A) and (B) are correct
   (4) Neither statement (A) nor (B) is correct

5. In the phenomenon of mirage, select an **incorrect** statement.
   (1) The refractive index of air increases with its density
   (2) Hotter air has smaller refractive index than the cooler air
   (3) If the air is still, the optical density at different layers of air increases with height
   (4) The apparent image of the object may create an illusion to the observer whether the temperature above the ground is non-uniform or uniform

6. Suppose while sitting in a parked car you notice a jogger approaching towards you in the side view mirror of R = 2 m. The jogger is running at a speed of 4 ms^{-1}. If the jogger is at a distance of 28 m from the mirror, then the average speed of the image of the jogger in one second is
   (1) $\frac{4}{725}$ ms^{-1} (2) $\frac{52}{675}$ ms^{-1}
   (3) $\frac{1}{150}$ ms^{-1} (4) Zero

7. Select the incorrect statement about the convex mirror.
   (1) It produces a virtual image, independent of the location of the real object
   (2) The ray, appearing to pass through the centre of curvature of it, retraces the path after the reflection
   (3) It can produce a real image
   (4) The virtual image of the real object, produced by it, is enlarged in size

8. The double-convex lenses are to be manufactured from a glass of refractive index 1.55, with the radius of curvature of one of the surfaces is double of the other. If the focal length is to be 20 cm then the radii of the lens is
   (1) 12 cm, 24 cm (2) 12.5 cm, 25 cm
   (3) 16 cm, 32 cm (4) 16.5 cm, 33 cm

9. A convex lens of focal length 15 cm is in contact with a concave lens of focal length 10 cm. A beam of light coming from infinity and falling on this combination, will
   (1) Diverge
   (2) Converge
   (3) Return to infinity
   (4) Remain undeviated

10. A 3.9 cm needle is placed 24 cm away from a convex mirror of focal length 15 cm. The size of the image of candle is
   (1) 1.5 cm (2) 2.5 cm
   (3) 4 cm (4) 4.5 cm

11. A prism of refracting angle 65° is made of glass of unknown refractive index. A parallel beam of light is incident on a face of the prism. The angle of minimum deviation is measured to be 55° in air. If the prism is placed in a liquid, then minimum deviation is 25°. The refractive index of the liquid is
   (1) $\frac{5}{3}$ (2) $\sqrt{\frac{4}{3}}$
   (3) $\sqrt{\frac{7}{3}}$ (4) $\sqrt{\frac{3}{2}}$

12. The image of a small electric bulb fixed on the wall of a room is to be obtained on the opposite wall 2 m away by means of a convex lens. The maximum possible focal length of the lens required for the purpose is
   (1) 0.30 m (2) 0.25 m
   (3) 0.75 m (4) 50 cm

13. A beam of light converges to a point P. A convex lens is placed in the path of the convergent beam 15 cm from P. If the focal length of the lens is 30 cm then the point where the beam converges, is
   (1) 15 cm from lens (2) 10 cm from P
   (3) 5 cm from P (4) 5 cm from lens

14. A small pin fixed on a table top is viewed from above from a distance of 40 cm. If it is viewed from the same point through a 12 cm thick glass slab of refractive index 1.5 held parallel to the table then the distance by which the pin appears to be raised up, is
   (1) 8 cm (2) 4 cm
   (3) 12 cm (4) 10 cm

15. A screen is placed 80 cm from an object. The image of the object on the screen is formed by a convex lens at two different locations separated by 16 cm. The focal length of the lens is nearly

(1) 19.2 m (2) 15.5 m

(3) 19.2 cm (4) 15.5 cm

16. Two thin prisms A and B made of glass of refractive index 1.6 and 1.5 respectively, are combined with each other so that this combination produces dispersion without deviation. If refracting angle of prism A is 10° then the refracting angle of second prism will be

(1) 12° (2) 16.6°

(3) 8.3° (4) 8°

17. The critical angle of flint glass and crown glass are θ_1 and θ_2 respectively. Then the correct relation is

(1) $\theta_1 > \theta_2$ (2) $\theta_1 < \theta_2$

(3) $\theta_1 = \theta_2$ (4) $\theta_1 = 2\theta_2$

18. A man with normal near point (25 cm) reads a book with small print using a thin convex lens of focal length 4 cm. The minimum angular magnification using the above simple microscope, is

(1) 6.25 (2) 7.25

(3) 6 (4) 5

19. A small telescope has an objective lens of focal length 120 cm and an eyepiece of focal length 4 cm. If the final image is formed at the least distance of distinct vision (25 cm). Then the magnifying power is

(1) 32 (2) 33.6

(3) 30 (4) 34.8

20. Read following statements **A** and **B** carefully and choose the option which represents the correct answer

Statement A: Mass density of an optically denser medium may be less than that of an optically rarer medium.

Statement B: Mass density of water is greater than that of turpentine but its optical density is lesser.

(1) Only statement (A) is correct

(2) Only statement (B) is correct

(3) Both the statements (A) and (B) are correct

(4) Neither statement (A) nor (B) is correct

21. Select incorrect option about light.

(1) In the wave picture of light, intensity of light is determined by the square of the amplitude of the wave

(2) When light travels from a rarer to a denser medium, the speed decreases

(3) When monochromatic light is incident on a surface separating two media, the reflected and refracted light both have the same frequency and wavelength

(4) For a given frequency, intensity of light in the photon picture, is defined by the number of photons crossing an unit area per unit time

22. Select the incorrect option about the variation in the interference fringes in YDSE.

(1) On increasing the distance of the screen from the plane of the slits, the angular separation of fringes remains constant

(2) On replacing the monochromatic source by a source of white light, the fringes closest on either side of the central white fringe is blue and the farthest will appear red

(3) On moving the source slit, closer to double-slit plane, The interference fringes may disappear

(4) On increasing the separation between two slits, the angular separation decreases

23. The distance between the two slits in double slit experiment is 2 mm. The width of each slit, to obtain 8 maxima of the double slit pattern within the central maxima of single slit pattern, is

(1) 0.5 mm (2) 0.2 mm

(3) 0.25 mm (4) 0.125 mm

24. Assume that the light of wavelength 5000 A° is coming from a star. If the objective of a telescope has a diameter of 2.5 m then the limit of resolution of the telescope is

(1) 2×10^{-7} rad (2) 4.12×10^{-7} rad

(3) 2.44×10^{-7} rad (4) 1.25×10^{-7} rad

25. Select the incorrect statement about the interference pattern with that seen for coherently illuminated single slit diffraction.
(1) The interference pattern has a number of equally spaced bright and dark bands
(2) The intensity falls as we go to successive maxima away from the centre, on either side
(3) The diffraction pattern is a superposition of a discontinuous family of waves originating from each point on a single slit
(4) The diffraction pattern has a central bright maximum which is twice as wide as the other maxima

26. In double slit experiment, using light of wavelength 500 nm, The angular width of a fringe formed on a distant screen is 0.2°. The spacing between the two slits is
(1) 3.4×10^{-4} m (2) 3×10^{-3} m
(3) 3×10^{-4} m (4) 1.4×10^{-4} m

27. If the aperture is 2 mm wide and the wavelength is 500 nm, then the Fresnel distance is
(1) 18 m (2) 8 m
(3) 6 m (4) 12 m

28. Two polaroids P_1 and P_3 are placed with their axis perpendicular to each other. An unpolarised light is incident on P_1. A third polaroid P_2 is kept in between P_1 and P_3 such that its axis makes on angle θ with that of P_1. The intensity of light emerging from P_3 will be maximum if θ is
(1) π (2) $\frac{\pi}{2}$
(3) $\frac{\pi}{4}$ (4) $\frac{\pi}{3}$

29. Select incorrect statement for the shape of the wavefront.
(1) Light, diverging from a point source, have spherical wavefront
(2) Light emerging out of a convex lens, when a point source is placed at its focus, have plane wavefront
(3) The portion of the wavefront of light from a distant star, intercepted by the Earth, is plane
(4) Light reflecting from a concave mirror, when a plane wave is incident on the concave mirror, have plane wavefront

30. In a single-slit diffraction. If the width of the slit is made half of the original width, then
(1) The size of the central diffraction band reduces to half of the initial size
(2) The intensity of the central diffraction band becomes four times
(3) The size of the central diffraction band becomes four times
(4) The intensity of the central diffraction band becomes one fourth of the initial

31. In a double slit experiment, when light of wavelength 500 nm was used, the angular width of a fringe was found to be 0.4° on a screen placed 1m away. If the entire experimental apparatus is immersed in water, then angular width of the fringe will be [$\mu_{water} = 4/3$]
(1) 0.1° (2) 0.05°
(3) 0.15° (4) 0.3°

32. Select incorrect statement about the speed of light in a medium.
(1) It is independent of the direction of propagation for isotropic media
(2) It is dependent on the motion of the source relative to the medium
(3) It does not depend on the nature of the source
(4) It is independent of intensity (in general)

33. Which phenomena is/are not possible for longitudinal wave?
(1) Diffraction (2) Refraction
(3) Polarization (4) Both (2) and (3)

34. The speed of light in glass for red and violet colour are v_1 and v_2 respectively. Then the correct relation between v_1 and v_2 is
(1) $v_1 > v_2$ (2) $v_2 > v_1$
(3) $v_1 = v_2$ (4) $v_1 = 2\,v_2$

35. Photoelectric effect in a photosensitive material of copper, is possible for
(1) Ultraviolet light (2) Green light
(3) Red light (4) All of these

36. Select incorrect statement about a photon-particle collision (such as photon-electron collision).
(1) Total energy is conserved
(2) Total momentum is conserved
(3) The number of photons are conserved
(4) Both (1) & (2)

37. Read following statements **A** and **B** carefully and choose the option which represents the correct answer

Statement A: In photoelectric effect, the stopping potential is independent of the intensity of incident radiation.

Statement B: The maximum kinetic energy of the photoelectrons varies nonlinearly with the frequency of incident radiation.

(1) Only statement (A) is correct
(2) Only statement (B) is correct
(3) Both the statements (A) & (B) are correct
(4) Neither statement (A) nor (B) is correct

38. Photoelectric current depends on
(1) The nature of the emitter material
(2) The intensity of incident light
(3) The potential difference applied between the two electrodes
(4) All of these

39. The maximum frequency of X-rays produced by 82.8 kV electrons is
(1) 1.5×10^{18} Hz (2) 6.2×10^{17} Hz
(3) 6×10^{18} Hz (4) 2×10^{19} Hz

40. The de-Broglie wavelength of the electrons accelerated through a potential difference of 64 V, is
(1) 6 Å (2) 1.5 Å
(3) 2 Å (4) 4.2 Å

41. An electron, an α-particle and a proton have the de-Broglie wavelength λ_e, λ_α and λ_P respectively. If kinetic energy is same for these particles, then the correct relation between wavelength is
(1) $\lambda_e < \lambda_\alpha < \lambda_P$ (2) $\lambda_e > \lambda_\alpha > \lambda_P$
(3) $\lambda_e > \lambda_P > \lambda_\alpha$ (4) $\lambda_e < \lambda_P < \lambda_\alpha$

42. Select the incorrect statement about photon.
(1) Photons are electrically neutral and are not deflected by electric and magnetic field
(2) On increasing the intensity of light of given wavelength, there is an increase in the number of photons per second crossing a given area, with each photon having the different energy
(3) In interaction of radiation with matter, radiation behaves as if it is made up of particles called photons
(4) Each photon has the speed of light

43. An electron has a wavelength of 3 nm. The momentum of the electron is
(1) 6.6×10^{-25} kg m/s (2) 3.3×10^{-24} kg m/s
(3) 2.2×10^{-25} kg m/s (4) 4.4×10^{-24} kg m/s

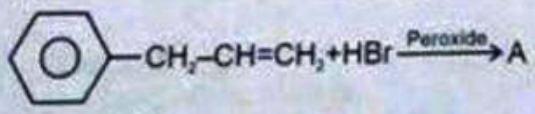
44. In an experiment on photoelectric effect, the slope of the cut-off voltage versus frequency of incident light, is found to be 4.14×10^{-15} V s. The value of Planck's constant is
(1) 6.59×10^{-34} J s (2) 6.60×10^{-34} J s
(3) 6.62×10^{-34} J s (4) 6.61×10^{-34} J s

45. The work function of a metal is 1 eV. When light of frequency 4×10^{14} Hz is incident on the metal surface, photoemission of electrons occurs. The maximum kinetic energy of the emitted electrons is
(1) 0.66 eV (2) 0.52 eV
(3) 1.25 eV (4) 1.5 eV

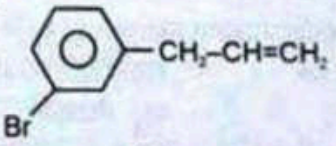
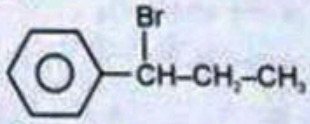
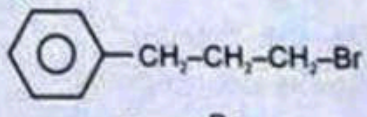
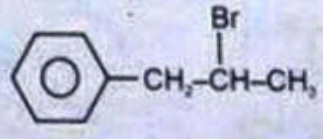
CHEMISTRY

46. Preferred reagent to replace hydroxyl group of alcohol by halogen is
(1) PBr_3 (2) PCl_5
(3) $SOCl_2$ (4) $NaBr/H_2SO_4$

47. Finkelstein reaction is used to prepare
(1) RF (2) RCl
(3) RBr (4) RI

48. 

Product A is

(1) 
(2) 
(3) 
(4) 

49. Correct order of boiling point of $CH_3CH_2CH_2CH_2Br$ (I), $CH_3CH_2\underset{\underset{Br}{|}}{C}HCH_3$ (II), and

$CH_3-\underset{\underset{Br}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_3$ (III)

(1) (I) > (II) > (III) (2) (II) > (I) > (III)

(3) (III) > (I) > (II) (4) (III) > (II) > (I)

50. $CH_3CH_2CH_2Cl \xrightarrow{KCN} A$, $CH_3CH_2CH_2Cl \xrightarrow{AgCN} B$

A and B are

(1) Chain isomers (2) Position isomers

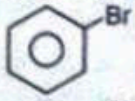
(3) Functional isomers (4) Identical

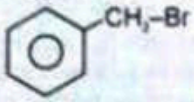
51. For the S_N2 reaction $RX + Nu \rightarrow R-Nu + X^-$, rate of reaction is maximum when R is

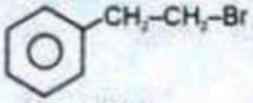
(1) Methyl (2) Ethyl

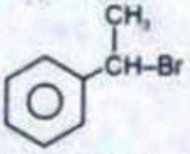
(3) Isopropyl (4) Tert-butyl

52. Maximum reactivity towards S_N1 reaction among the following is of

(1) 

(2) 

(3) 

(4) 

53. Chiral molecule among the following is

(1) Butan-2-ol (2) Butan-1-ol

(3) Butane (4) Butene

54. Major product of the reaction is

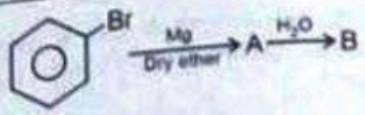
$CH_3-CH_2-CH_2-\underset{\underset{Br}{|}}{CH}-CH_3 \xrightarrow{\text{alc. KOH}}$

(1) $CH_3-CH_2-CH_2-CH=CH_2$

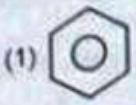
(2) $CH_3-CH=CH-CH_2-CH_3$

(3) $CH_3-CH_2-\underset{\underset{CH_3}{|}}{C}=CH_2$

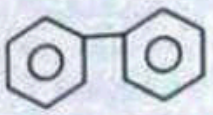
(4) $CH_2-\underset{\underset{CH_3}{|}}{CH}-CH=CH_2$

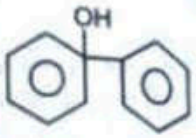
55. 

Product B is

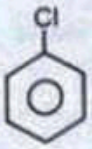
(1) 

(2) 

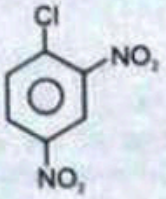
(3) 

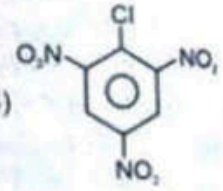
(4) 

56. Maximum reactivity towards NaOH is of

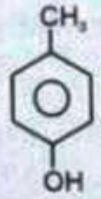
(1) 

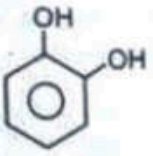
(2) 

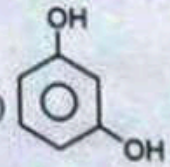
(3) 

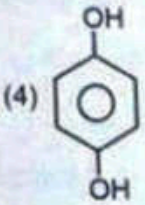
(4) 

57. Hydroquinone is

(1) 

(2) 

(3) 

(4) 

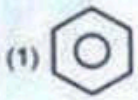
58. Which of the following on reaction with Grignard reagent gives primary alcohol?

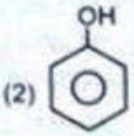
(1) HCHO (2) CH_3CHO

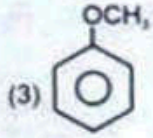
(3) CH_3COCH_3 (4) CH_3COOH

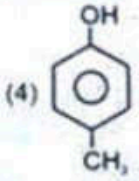
59. 

Product A is

(1) 

(2) 

(3) 

(4) 

60. Most acidic among the following is

(1) p-Nitrophenol (2) o-Nitrophenol

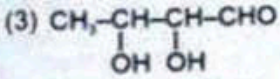
(3) p-Cresol (4) Phenol

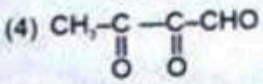
61. $CH_3-CH=CH-CH_2-OH \xrightarrow{PCC} A$

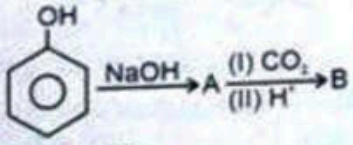
Product A is

(1) $CH_3-CH=CH-CHO$

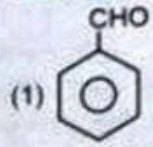
(2) $CH_3-CH=CH-COOH$

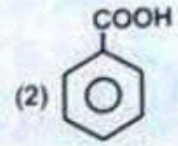
(3) 

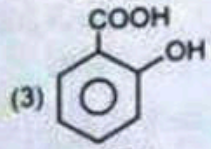
(4) 

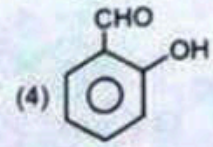
62. 

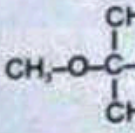
Product B is

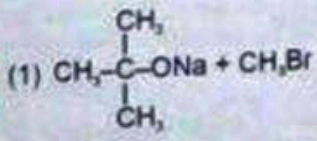
(1) 

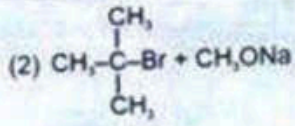
(2) 

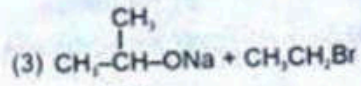
(3) 

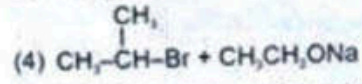
(4) 

63.  can be prepared by the reaction of

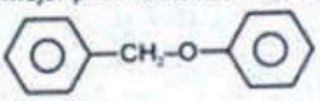
(1) 

(2) 

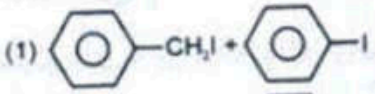
(3) 

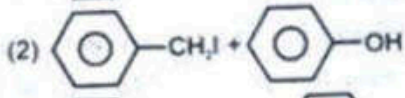
(4) 

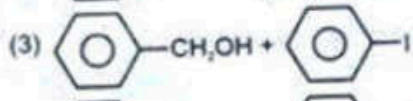
64. Major product that are formed by heating

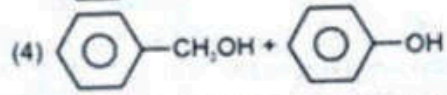


with HI are

(1) 

(2) 

(3) 

(4) 

65. Denaturation of alcohol is done by the help of

(1) $CuSO_4$ (2) $ZnSO_4$

(3) H_2SO_4 (4) HCl

66. $RCN + SnCl_2 + HCl \rightarrow RCH=NH \xrightarrow{H_3O^+} RCHO$

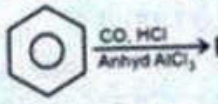
Above reaction is known as

(1) Stephen reduction

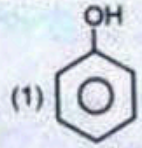
(2) Rosenmund reduction

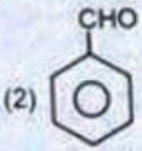
(3) Wolff-Kishner reduction

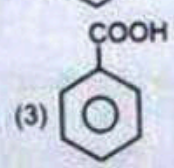
(4) Clemmensen reduction

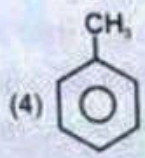
67. 

Product P in the reaction is

(1) 

(2) 

(3) 

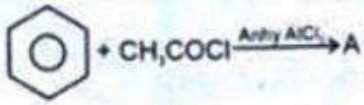
(4) 

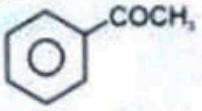
68. $CH_3-C\equiv CH \xrightarrow[(II)\ NaOH/H_2O_2]{(I)\ B_2H_6} A$

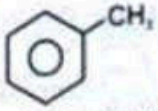
A is

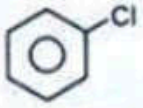
(1) Propanol (2) Propanone

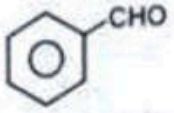
(3) Propanal (4) Propanoic acid

69. 

(1) 

(2) 

(3) 

(4) 

70. Schiff's base is formed by the reaction of aldehyde with

(1) Ammonia (2) Amines

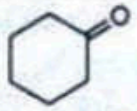
(3) Hydrazine (4) Phenyl hydrazine

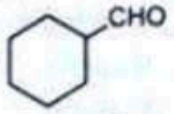
71. Ion used in Tollen's reagent is

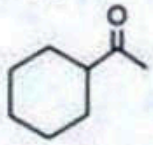
(1) Ag^+ (2) $[Ag(NH_3)_2]^+$

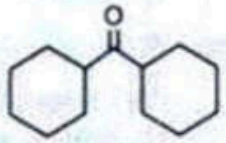
(3) Cu^{2+} (4) Cu^+

72. Haloform reaction is shown by

(1) 

(2) 

(3) 

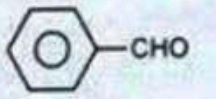
(4) 

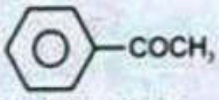
73. Which of the following forms 2,4-DNP derivative and does not reduce Tollen's or Fehling reagent?

(1) HCHO (2) CH_3CHO

(3) CH_3COCH_3 (4) HCOOH

74. Which of the following can undergo aldol condensation?

(1)  (2) $(CH_3)_2CH-CHO$

(3)  (4) $(CH_3)_3C-CHO$

75. $HCHO \xrightarrow[KOH,\ \Delta]{conc} A + B$

Product A and B are

(1) CH_3OH and HCOOH

(2) CH_3OK and HCOOH

(3) CH_3OH and HCOOK

(4) CH_3OK and HCOOK

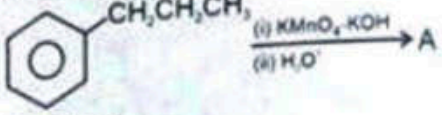
76. Malonic acid is

(1) HOOC–COOH

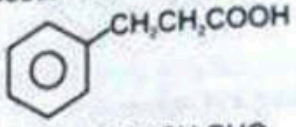
(2) $HOOC-CH_2-COOH$

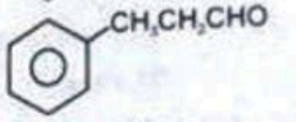
(3) $HOOC-(CH_2)_2-COOH$

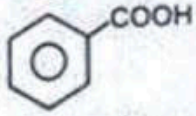
(4) $HOOC-(CH_2)_3-COOH$

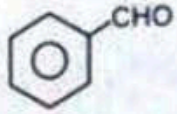
77. 

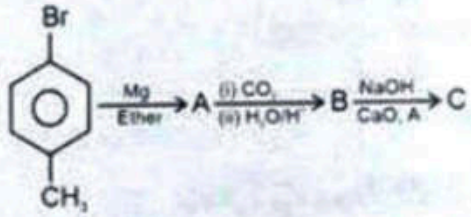
Product A is

(1) 

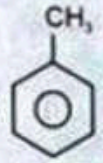
(2) 

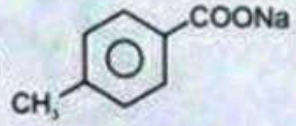
(3) 

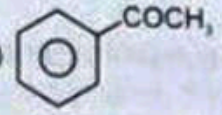
(4) 

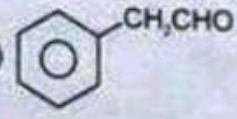
78. 

Product C is

(1) 

(2) 

(3) 

(4) 

79. Minimum pKa among the following is of

(1) CF_3COOH (2) NO_2CH_2COOH

(3) $NC-CH_2-COOH$ (4) CH_3CH_2COOH

80. $CH_3CH_2COOH \xrightarrow[(ii)\ H_2O]{(i)\ Br_2/Red\ P} A$

Product A in the above reaction is

(1) CH_3CH_2COBr (2) $CH_2(Br)-CH_2-COOH$

(3) $CH_3-CH(Br)-COOH$ (4) $CH_3-CH(Br)-COBr$

81. Maximum reactivity towards nucleophilic addition reaction is of

(1) HCHO (2) CH_3CHO

(3) CH_3COCH_3 (4) CH_3CH_2CHO

82. Which alkene on ozonolysis gives ethanal and propanone?

(1) $CH_3-CH=CH_2$ (2) $CH_3-CH=CH_2-CH_3$

(3) $CH_3-C(CH_3)=CH-CH_3$ (4) $CH_3-C(CH_3)=C(CH_3)-CH_3$

83. Identify the one that on photochemical chlorination yields three isomeric monochlorides (excluding stereoisomers)

(1) CH_3CH_3

(2) $CH_3CH_2CH_3$

(3) $CH_3CH_2CH_2CH_3$

(4) $CH_3CH_2CH_2CH_2CH_3$

84. Chloroform slowly oxidised by air in presence of light to form

(1) CO (2) $COCl_2$

(3) CO_2 (4) HCHO

85. Ethyl alcohol on reaction with H_2SO_4 at 413 K gives which of the following main product?

(1) Alkene (2) Alkane

(3) Ether (4) Aldehyde

86. $(CH_3)_2C=O \xrightarrow{NH_2NH_2} A \xrightarrow[Ethylene\ glycol]{KOH} B + N_2$

Product B is

(1) $CH_3CH_2CH_3$ (2) $CH_3CH_2CH_2OH$

(3) $CH_3CH(OH)CH_3$ (4) $CH_3CH_2OCH_3$

87. $CH_3COOH \xrightarrow[(ii)\ H_3O^+]{(i)\ LiAlH_4} A \xrightarrow{PCl_3} B \xrightarrow[Dry\ ether]{Na} C$

Product C is

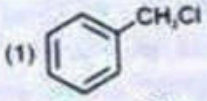
(1) CH_3CH_3

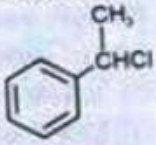
(2) $CH_3CH_2CH_3$

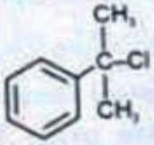
(3) $CH_3CH_2CH_2CH_3$

(4) $CH_3CH_2CH_2CH_2CH_3$

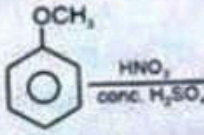
88. Which of the following is/are benzylic chloride?

(1) 

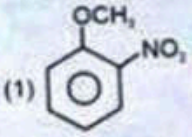
(2) 

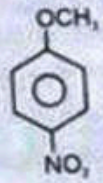
(3) 

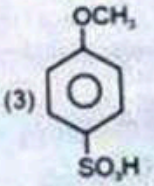
(4) All of these

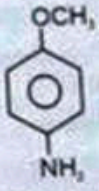
89. 

(major)

(1) 

(2) 

(3) 

(4) 

90. Ethanoic acid on treatment with P_2O_5 gives

(1) Ester (2) Anhydride

(3) Aldehyde (4) Alcohol

BOTANY

91. Reproduction is synonymous with growth in all of the given organisms, **except**

(1) Bacteria (2) *Amoeba*

(3) Unicellular Algae (4) Most Fungi

92. Indian Botanical Garden is located in which of the given city of India?

(1) Howrah (2) Lucknow

(3) New Delhi (4) Chennai

93. Which of the given groups of bacteria are present in the guts of several ruminant animals such as cows and buffaloes and are responsible for production of methane (biogas) from dung of these animals?

(1) Halophiles

(2) Thermoacidophiles

(3) Methanogens

(4) Eubacteria

94. Which of the given protists has a protein rich layer called pellicle instead of a cell wall?

(1) Dinoflagellates (2) Euglenoids

(3) Slime moulds (4) Diatoms

95. Artificial system of classification was based on

(1) Natural affinities among the organisms

(2) Evolutionary relationships between various organisms

(3) Phytochemistry

(4) Vegetative and sexual characteristics

96. Which of the given plant groups includes horsetails and ferns?

(1) Algae (2) Bryophyte

(3) Pteridophyte (4) Gymnosperm

97. Cells at region of meristematic activity in root-tip exhibits all of the given properties, **except**

(1) Are thin-walled

(2) Have dense protoplasm

(3) Gradually differentiate and mature

(4) Divide repeatedly

98. Which of the given is a medicinal plant of family Fabaceae?

(1) Muliathi (2) Belladonna

(3) Ashwagandha (4) *Aloe*

99. Select the **incorrect** statement w.r.t. 'trichomes'.

(1) Are usually unicellular in the shoot system

(2) May be branched or unbranched

(3) May even be secretory

(4) Helps in preventing water loss due to transpiration

100. Read the statements, stating them true (T) or false (F) and select the **correct** option.

A Annual rings seen in a cut stem give an estimate of the age of the tree.

B Spring wood is lighter in colour and has a lower density.

C Autumn wood is darker and has a higher density.

D Lenticels permit the exchange of gases between the outer atmosphere and the internal tissue of the stem.

	A	B	C	D
(1)	F	F	F	T
(2)	F	T	F	F
(3)	T	F	F	T
(4)	T	T	T	T

101. Which of the given chromosomes has middle centromere forming two equal arms of the chromosome?

(1) Metacentric chromosome

(2) Sub-metacentric chromosome

(3) Acrocentric chromosome

(4) Telocentric chromosome

102. Which of the given cell organelles is formed by the process of packaging in the Golgi apparatus?

(1) Vacuole

(2) Endoplasmic reticulum

(3) Lysosome

(4) Ribosome

103. Read the given statements and select the **correct** option.

Statement A: Cell growth results in disturbing the ratio between the nucleus and the cytoplasm.

Statement B: Mitotic divisions in the meristematic tissues-the apical and the lateral cambium, result in a continuous growth of plants throughout their life.

(1) Only statement A is correct

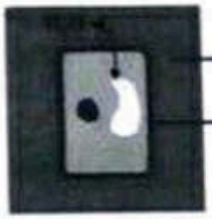
(2) Only statement B is correct

(3) Both the statements are correct

(4) Both the statements are incorrect

104. The recombinase-mediated process occurs in which of the given stages of prophase I of meiosis?
(1) Leptotene (2) Zygotene
(3) Pachytene (4) Diplotene

105. Select the **incorrect** statement w.r.t. the given diagram



(1) There is no net flow of water towards the inside or outside the plant cell
(2) The external solution balances the osmotic pressure of the cytoplasm of the plant cell
(3) Water flows into the cell and out of the cell are in equilibrium
(4) The plant cell is tending towards turgid state

106. The bulk movement of substances through the conducting or vascular tissues of plants is called
(1) Translocation (2) Transpiration
(3) Imbibition (4) Guttation

107. Nitrogen is the mineral element required by the plants in the greatest amount. It is absorbed mainly as
(1) NO_2^- (2) NO_3^-
(3) NH_4^+ (4) Both (1) and (2)

108. Read the given statements, stating them true (T) or, false (F) and select the **correct** option.
A In legume plants, division and growth of cortical and pericycle cells lead to nodule formation.
B Nitrogen is a limiting nutrient for both natural and agricultural ecosystem.
C One of the toxic effects of manganese in plants is inhibition of calcium translocation in shoot apex.
D Some elements like N, S, Mo delay flowering if their concentration in plants is low.

	A	B	C	D
(1)	T	T	T	T
(2)	T	T	F	T
(3)	F	F	T	T
(4)	F	F	F	F

109. For every CO_2 molecule entering the Calvin cycle, how many molecules of ATP and NADPH respectively are required?
(1) 3 and 2 (2) 2 and 3
(3) 1 and 2 (4) 2 and 1

110. The primary CO_2 acceptor w.r.t. C_4 cycle is
(1) Phosphoenol pyruvate
(2) Ribulose bisphosphate
(3) Oxaloacetic acid
(4) Phosphoglyceric acid

111. In which of the given stages of glycolysis, ATP is utilised?
A Glucose → Glucose-6-phosphate
B Glucose-6-phosphate → Fructose-6-phosphate
C Fructose-6-phosphate → Fructose-1,6-bisphosphate
D Fructose-1,6-bisphosphate → Glyceraldehyde-3-phosphate
(1) A and B (2) B and C
(3) A and C (4) B and D

112. When proteins are respiratory substrate, the ratio of volume of CO_2 evolved to the volume of O_2 consumed would be about
(1) 1.0 (2) 0.7
(3) 1.33 (4) 0.9

113. In most situations, which of the given plant hormones acts as an antagonist to GAs?
(1) Auxin (2) ABA
(3) Cytokinin (4) Ethylene

114. Match the Column-I and Column-II w.r.t. to plant hormones and their applications.

	Column-I	Column-II
a	Auxin	(i) Helps seed to withstand desiccation
b	Cytokinin	(ii) Induces parthenocarpy in tomatoes
c	Ethylene	(iii) Helps to produce new chloroplasts in leaves
d	Abscisic acid	(iv) Induces flowering in mango

	a	b	c	d
(1)	(ii)	(iii)	(iv)	(i)
(2)	(i)	(ii)	(iii)	(iv)
(3)	(iv)	(iii)	(ii)	(i)
(4)	(iii)	(iv)	(i)	(ii)

115. Select the **incorrect** match w.r.t. organisms and their asexual reproductive structures.
(1) *Penicillium* - Conidia
(2) *Chlamydomonas* - Zoospore
(3) Yeast - Bud
(4) *Agave* - Offset

116. Select the **incorrect** statement w.r.t. syngamy.
(1) It is the most vital event of sexual reproduction
(2) Includes fusion of gametes
(3) The term fertilisation is also often used for syngamy
(4) Occurs inside the body of majority of algae and fishes as well as amphibians

117. Which of the given layers of anther possesses dense cytoplasm and generally have more than one nucleus?
(1) Epidermis (2) Middle layer
(3) Endothecium (4) Tapetum

118. In flowering plants, the body of the ovule fuses with funicle in the region called
(1) Cotyledon (2) Hilum
(3) Micropyle (4) Integument

119. Which of the following crosses in plant is considered to be test cross?
(1) Yellow seeded plant × Green seeded plant
(2) Violet flowered plant × Violet flowered plant
(3) Full pod plant × White flowered plant
(4) Dwarf plant × Green pod plant

120. Select the **incorrect** statement w.r.t. sickle-cell anaemia.
(1) It is an autosomal linked recessive trait
(2) It can be transmitted from parents to offspring when both the partners are carrier for that particular gene (or heterozygous)
(3) It is controlled by a single pair of allele
(4) It is caused by the substitution of valine (val) by Glutamic acid (Glu) at the sixth position of the beta globin chain of the haemoglobin molecule

121. One codon codes for only one amino acid. It reflects which of the given features of genetic code?
(1) Commaless (2) Unambiguous
(3) Degenerate (4) Universality

122. DNA as an acidic substance present in the nucleus was first identified by
(1) Erwin Chargaff (2) Maurice Wilkin
(3) Friedrich Miescher (4) Rosalind Franklin

123. All of the given diseases are caused by fungi, **except**
(1) Brown rust of wheat
(2) Black rot of crucifers
(3) Red rot of sugarcane
(4) Late blight of potato

124. Which of the following should be performed to obtain virus-free plants from diseased plant?
(1) Growing of meristems *in vitro*
(2) Somatic hybridisation
(3) Fusion of protoplasts from two different plants
(4) Growing of embryo *in vitro*

125. Full potential of penicillin as an effective antibiotic was established by
(1) Alexander Fleming (2) Ernest Chain
(3) Howard Florey (4) Both (2) and (3)

126. Which of the given bioactive molecule acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol in the human body?
(1) Statin (2) Cyclosporin-A
(3) Lipase (4) Streptokinase

127. Which of the given population attributes cause decline in population density?
(1) Natality and Mortality
(2) Mortality and Emigration
(3) Immigration and Emigration
(4) Natality and Immigration

128. The interaction where one species is benefitted and the other is neither benefitted nor harmed is called

(1) Commensalism (2) Amensalism

(3) Competition (4) Mutualism

129. Read the following statements and select the **correct** option

Statement A: Pyramid of energy is always upright.

Statement B: Transfer of energy at successive trophic level follows 10 percent law.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements are correct

(4) Both the statements are incorrect

130. Pioneer community in hydrarch succession is

(1) Phytoplankton

(2) Free-floating angiosperm

(3) Rooted hydrophytes

(4) Forest

131. Which of the given vertebrates contribute to maximum global biodiversity?

(1) Fishes (2) Mammals

(3) Birds (4) Reptiles

132. The most important cause driving animals and plants to extinction is

(1) Habitat loss and fragmentation

(2) Over-exploitation

(3) Alien species invasions

(4) Co-extinctions

133. Which of the given green house gases w.r.t. their relative contribution, contributes maximum to total global warming?

(1) N_2O (2) CFC

(3) CH_4 (4) CO_2

134. Match Column-I with Column-II and select the **correct** option

	Column-I	**Column-II**
a	Ahmed Khan	(i) Created Haryana Kisan Welfare Club
b	Amrita Devi Bishnoi	(ii) Remedy to plastic waste
c	Ramesh Chandra Dagar	(iii) Sacrificed life for saving trees

	a	b	c
(1)	(i)	(ii)	(iii)
(2)	(ii)	(iii)	(i)
(3)	(iii)	(ii)	(i)
(4)	(i)	(iii)	(ii)

135. The sequence of chromosome 1 w.r.t. Human Genome Project was completed only in

(1) May, 2006 (2) August, 2003

(3) July, 2005 (4) May, 2003

ZOOLOGY

136. Read the following statements **A** and **B** carefully and choose the option which represents the **correct** answer

Statement A: Reproduction unables the continuity of the species generation after generation.

Statement B: Zygote is the vital link that ensures continuity of species between organisms of one generation.

(1) Both the statements A and B are incorrect

(2) Both the statements A and B are correct

(3) Only statement A is correct

(4) Only statement B is correct

137. Sperm of Rat contains 21 chromosomes. What will be number of chromosomes in their meiocytes?

(1) 21 (2) 42

(3) 63 (4) 84

138. All of the following are differences between male and female reproductive events/structures **except**

(1) In female, Oogenesis starts during her foetal life whereas in male it starts at puberty

(2) Sperm formation continues even in old men, but formation of ovum ceases in women around the age of 50 years

(3) Four sperms are produced from a primary spermatocyte and single ootid is produced from a primary oocyte

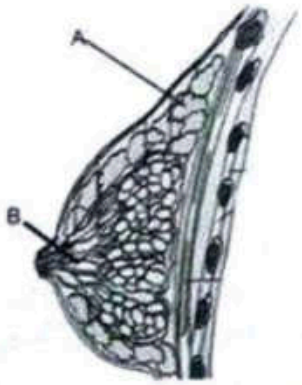
(4) Male germ cells are haploid whereas oogonia in females are diploid

139. Seminal plasma does not contains

(1) Enzyme (2) Glycogen

(3) Ca^{+2} (3) Fructose

140. Diagrammatic sectional view of mammary gland is shown in the following figure. Choose the option which **correctly** represents A and B



S. No	A	B
1.	Mammary lobe	Lactiferous duct
2.	Mammary alveolus	Mammary duct
3.	Fat	Ampulla
4.	Mammary duct	Areola

141. During embryonic development, certain cells called stem cells have the potency to give rise to all the tissues and organs. Such stem cells are present in

(1) Trophoblast

(2) Outer cell mass

(3) Inner cell mass

(4) Ectoderm

142. Rapid decline in all of the following is responsible for increase in population growth **except**

(1) Death rate

(2) Maternal mortality rate

(3) Number of people in reproducible age

(4) Infant mortality rate

143. Select the **correct** statement w.r.t. steroidal oral contraceptive pills.

(1) They contain small doses of either progesterone or oestrogen

(2) Combined pills have to be taken daily throughout menstrual cycle starting preferably within the first five days of menstrual cycle

(3) They inhibit ovulation and implantation but not alter quality of cervical mucus

(4) Pills are very effective and well accepted by females

144. Select the **incorrect** statement w.r.t. assisted reproductive technologies.

(1) Test tube baby programme includes both IVF and ET

(2) Embryo formed either by *in vitro* fertilisation or by *in vivo* fertilisation can be used for IUT

(3) In ICSI, a sperm is directly injected into the ovum within fallopian tube

(4) Only *in vivo* fertilisation is possible in GIFT

145. Leghorn is breed of **A** and Jersey is breed of **B**. Choose the option which **correctly** represents **A** and **B**

S. No.	A	B
1.	Cartilaginous fishes	Goat
2.	Bony fishes	Sheep
3.	Chickens	Cow
4.	Cow	Chicken

146. All of the following are fresh water edible fishes **except**

(1) Catla (2) Rohu

(3) Common carp (4) Pomfrets

147. In stirring type bioreactors, the stirrer facilitates

(1) Oxygen availability (2) Product removal

(3) pH control (4) Regulation of process

148. In rDNA technology, rDNA means

(1) DNA with a piece of RNA

(2) DNA with a piece of foreign DNA

(3) DNA which takes part in recombination

(4) DNA not associated with foreign gene

149. The genetic defect adenosine deaminase(ADA) deficiency may be cured permanently by
(1) Administering adenosine deaminase promoters
(2) Introducing bone marrow cells producing enzyme only once
(3) Enzyme replacement therapy
(4) Isolating genes from marrow cells producing ADA and introducing those genes into cells at early embryonic stages

150. Which of the following is **incorrect** statement for Bt toxin?
(1) Crystals of Bt toxin produced by some bacteria kill the bacteria themselves
(2) *Cry I Ab* is effective against corn borer
(3) Bt toxin is not harmful to mammals
(4) *Cry I Ac* is effective against cotton bollworms

151. Birds with average sized wings, survived in a severe storm more successfully than with longer or shorter wings. It illustrates
(1) Stabilising selection
(2) Gene flow
(3) Disruptive selection
(4) Founder effect

152. If the frequency of a dominant allele 'B' in a given population is 0.64, then find out the frequency of other recessive allele 'b' in the same population when population is in Hardy-Weinberg equilibrium.
(1) 0.064 (2) 0.6
(3) 0.36 (4) 0.96

153. Which species of *Homo* used hides to protect their body and buried their dead?
(1) *Homoerectus*
(2) *Homo habilis*
(3) Neanderthal man
(4) First human-like being the hominid

154. Which of the following animals is **not** evolved from thecodonts?
(1) Crocodiles (2) Dinosaurs
(3) Birds (4) Tuataras

155. Select the **mismatch** w.r.t. disease and its causative organism.
(1) Typhoid – *Salmonella typhi*
(2) Pneumonia – *Haemophilous influenzae*
(3) Malaria – *Plasmodium falciparum*
(4) Amoebiasis – *Wuchereria bancrofti*

156. Opioid receptors are present in our
(1) Gastrointestinal tract
(2) Peripheral neural system
(3) Both GIT and PNS
(4) Both CNS and ANS

157. Read the following statements carefully and choose the option containing all **incorrect** statements.
(a) Skin on our body is main physical barrier
(b) Macrophages and neutrophils in blood can phagocytize and destroy microbes
(c) Interferons protect virus-infected cells from further viral infections
(d) Saliva in mouth and tears in eyes prevents microbial growth
(1) a and b (2) b and c
(3) c and d (4) a and d

158. Radial symmetry is **absent** in members of phylum
(1) Coelenterata (2) Ctenophora
(3) Echinodermata (4) Annelida

159. Notochord is mesodermal rod like structure present
(1) Dorsal to nerve cord in larval stage of urochordates only
(2) Dorsal to nerve cord only in adult of cephalochordates
(3) Ventral to nerve cord in adult of only urochordates
(4) Ventral to nerve cord in embryo of all chordates

160. Proboscis gland is an excretory organ in
(1) *Echinus*
(2) *Balanoglossus*
(3) *Branchiostoma*
(4) *Chaetopleura*

161. Life cycle of *Obelia* represents alternation of generation known as
(1) Metastasis
(2) Metamorphosis
(3) Metagenesis
(4) Mutation

162. Select the **incorrect** match w.r.t. female cockroach.
(1) Collaterial glands – One pair
(2) Spiracles – 10 pairs
(3) Ovarioles – 8 pairs
(4) Anal style – One pair

163. Match the Column-I with Column-II and choose the option with all **correct** match

Column-I	Column-II
(a) Connective tissue	(i) Constitutes 40% - 50% of human body weight
(b) Epithelial tissue	(ii) Most abundant and widely distributed in the body
(c) Muscular tissue	(iii) Cells are compactly packed with little intercellular matrix
(d) Neutral tissue	(iv) Exert the greatest intercellular control over the body's responsiveness to changing conditions

(1) (a)-ii, (b)-iii,(c)-i,(d)-iv
(2) (a)-i, (b)-ii, (c)-iii, (d)-iv
(3) (a)-ii, b-i, c-iii, d-iv
(4) (a)-iv, (b)-iii, (c)-ii, (d)-i

164. Acid insoluble fraction called retentate contains all polymeric compounds on chemical analysis of living tissue **except**
(1) Proteins
(2) polysaccharides
(3) Nucleic acids
(4) Fatty acids

165. Read the following statements w.r.t. enzymes and choose the option with all **correct** statements.
(a) Difference in average energy content of substrate and product is called activation energy
(b) Substrate binds to the active site of product to form substrate enzyme complex
(c) Activity of enzyme declines both above and below the optimum value of temperature and pH
(d) Competitive inhibitors are often used in the control of bacterial pathogens
(1) a and b (2) b and c
(3) c and d (4) a and d

166. How many of the following given statements are **true**?
(a) Gross calorific value of carbohydrates is less than the physiological value of proteins
(b) Gross calorific value of proteins is more than the physiological value of carbohydrates
(c) Gross calorific value of carbohydrates is equal to the gross calorific value of proteins
(d) Physiological value of Carbohydrates is equal to the physiological value of proteins
(1) 3 (2) 2
(3) 4 (4) 1

167. Incomplete cartilaginous rings are **not** found in
(1) Trachea
(2) Bronchi
(3) Terminal bronchiole
(4) Initial bronchiole

168. The amount CO_2 delivered to alveoli by deoxygenated blood in each cardiac output is approximately
(1) 4 ml (2) 40 ml
(3) 200 ml (4) 400 ml

169. Megakaryocytes are present in
(1) Thymus
(2) Spleen
(3) Bone marrow
(4) Lymph modes

170. Which of the following layers of blood vessels contain connective tissue fibres?
(1) Tunica externa only
(2) Tunica intima only
(3) Tunica media only
(4) All layers

171. Select the **correct** match w.r.t. animals and their excretory products.
(1) Aquatic insects – Ureotelic
(2) Aquatic amphibians – Ammonotelic
(3) Mammals – Uricotelic
(4) Land snails – Ureotelic

172. Which of the following is **not** a part of nephron?
(1) Glomerulus
(2) PCT
(3) DCT
(4) Collecting duct

173. Choose the option which fills the blank **correctly** to complete the analogy
Amoeboid movement : Macrophages :: Flagellar movement : _____.
(1) *Euglena* (2) *Amoeba*
(3) *Paramoecium* (4) *Leucocytes*

VIP HAWELI

174. In organised skeletal muscle, muscle bundles held together by a common collagenous connective tissue layer called
(1) Fascia
(2) Sarcolemma
(3) Intercalated disc
(4) Fascicles of hypothalamus

175. The hormones which regulate gene expression by the interaction of hormone receptor complex with the genome are
(1) Iodothyronines
(2) Catecholamines
(3) Releasing hormones of hypothalamus
(4) Hormones of gastrointestinal tract

176. Which of the following organs is not helpful in erythropoiesis?
(1) Adrenal gland (2) Thymus
(3) Kidney (4) Thyroid gland

177. In organ of corti, nerve impulses are generated due to bending of sensory hair cells in associated A and are analysed and sound is recognised by B. A and B respectively represents
(1) Motor neurons basilar membrane
(2) Sensory nerve fibres, organ of corti
(3) Organ of corti, olfactory cortex
(4) Sensory nerve fibres, auditory cortex

178. Which of the following is not under control of medulla oblongata?
(1) Gastric secretion
(2) Cardiovascular reflexes
(3) Respiration
(4) Urge of eating

179. Select the **incorrect** match
(1) Acromegaly – Hypersecretion of GH
(2) Grave's disease – Hypothyroidism
(3) Diabetes insipidus – Hyposecretion of ADH
(4) Addison's disease – Hypoadrenocorticism

180. Match the Column-I with Column-II and Choose the option with only **correct** match

Column-I	Column-II
(a) Smooth muscle	(i) Myoglobin
(b) Tropomyosin	(ii) Thin filament
(c) Red muscle	(iii) Sutures
(d) Skull	(iv) Involuntary

(1) (a)-i, (b)-ii,(c)-iii,(d)-iv
(2) (a)-ii, (b)-iii, (c)-i, (d)-iv
(3) (a)-(iv), (b)-ii, (c)-i, (d)-iii
(4) (a)-iv, (b)-iii, (c)-ii, (d)-i

❑❑❑

Aakash
Medical | IIT-JEE | Foundations
Regd. Office :Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

NCERT BOOSTER TEST SERIES

Test – 09 (Answer Key)

1.	(3)	40.	(2)	79.	(1)	118.	(2)	154.	(4)
2.	(4)	41.	(3)	80.	(3)	119.	(1)	155.	(4)
3.	(2)	42.	(2)	81.	(1)	120.	(4)	156.	(1)
4.	(3)	43.	(3)	82.	(3)	121.	(2)	157.	(2)
5.	(4)	44.	(3)	83.	(4)	122.	(3)	158.	(4)
6.	(1)	45.	(1)	84.	(2)	123.	(2)	159.	(4)
7.	(4)	46.	(3)	85.	(3)	124.	(1)	160.	(2)
8.	(4)	47.	(4)	86.	(1)	125.	(4)	161.	(3)
9.	(1)	48.	(3)	87.	(3)	126.	(1)	162.	(4)
10.	(1)	49.	(1)	88.	(4)	127.	(2)	163.	(1)
11.	(4)	50.	(3)	89.	(2)	128.	(1)	164.	(4)
12.	(4)	51.	(1)	90.	(2)	129.	(3)	165.	(3)
13.	(3)	52.	(4)	91.	(4)	130.	(1)	166.	(2)
14.	(2)	53.	(1)	92.	(1)	131.	(1)	167.	(3)
15.	(3)	54.	(2)	93.	(3)	132.	(1)	168.	(3)
16.	(1)	55.	(1)	94.	(2)	133.	(4)	169.	(3)
17.	(2)	56.	(4)	95.	(4)	134.	(2)	170.	(4)
18.	(1)	57.	(4)	96.	(3)	135.	(1)	171.	(2)
19.	(4)	58.	(1)	97.	(3)	136.	(1)	172.	(4)
20.	(3)	59.	(2)	98.	(1)	137.	(2)	173.	(1)
21.	(3)	60.	(1)	99.	(1)	138.	(4)	174.	(1)
22.	(2)	61.	(1)	100.	(4)	139.	(2)	175.	(1)
23.	(1)	62.	(3)	101.	(1)	140.	(3)	176.	(2)
24.	(3)	63.	(1)	102.	(3)	141.	(3)	177.	(4)
25.	(3)	64.	(2)	103.	(3)	142.	(3)	178.	(4)
26.	(4)	65.	(1)	104.	(3)	143.	(4)	179.	(2)
27.	(2)	66.	(1)	105.	(4)	144.	(3)	180.	(3)
28.	(3)	67.	(2)	106.	(1)	145.	(3)		
29.	(4)	68.	(3)	107.	(2)	146.	(4)		
30.	(4)	69.	(1)	108.	(1)	147.	(1)		
31.	(4)	70.	(2)	109.	(1)	148.	(2)		
32.	(2)	71.	(2)	110.	(1)	149.	(4)		
33.	(3)	72.	(3)	111.	(3)	150.	(1)		
34.	(1)	73.	(3)	112.	(4)	151.	(1)		
35.	(1)	74.	(3)	113.	(2)	152.	(3)		
36.	(3)	75.	(3)	114.	(1)	153.	(3)		
37.	(1)	76.	(2)	115.	(4)				
38.	(4)	77.	(3)	116.	(4)				
39.	(4)	78.	(1)	117.	(4)				

1 of 1